IEEE WIRELESS COMMUNICATIONS LETTERS

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Journal:	*IEEE Wireless Communications Letters*
Manuscript ID	WCL2026-0016
Manuscript Type:	Original Article
Date Submitted by the Author:	02-Jan-2026
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Key Words:	Propagation < Antennas and Propagation, MIMO systems < Transmission Technology, Cellular technology < Wireless Personal Communication Systems

Secrecy Sum Rate for NOMA-Assisted Downlink Pinching-Antenna Systems

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***Abstract*—Non-orthogonal multiple access (NOMA) technology can be applied to pinching-antenna systems (PASS) to enhance the utilization efficiency of spectrum resources and communication confidentiality. This paper investigates a NOMA assisted downlink PASS, which flexibly serves users by dynamically activating pinching antenna (PA) along a dielectric waveguide. To adress the security challenges induced by the broadcasting nature of the NOMA-PASS integration, our primary objective is to maximize the secrecy sum rate subject to legitimate achievable rate constraints. The resulting problem non-convex requires the joint optimization of transmit power and pinch positions. To tackle this challenge we employ an alternating optimization (AO) framework. Specifically, successive convex approximation (SCA) is adopted for power allocation, while the particle swarm optimization (PSO) algorithm is employed to address the pinch positioning subproblem. Simulation results confirm that the proposed scheme significantly surpasses conventional fixed-antenna systems in terms of secrecy sum rate performance.**

***Index Terms*—non-orthogonal multiple access, pinching-antenna systems, secrecy sum-rate maximization.**

I. Introduction

As the evolution of sixth-generation (6G) wireless communications continue to evolve, communication systems, while pursuing higher data rates and improved reliability, face significant challenges from complex, dynamic, and unpredictable wireless propagation environments. Consequently, flexible antenna technologies, which electromagnetic propagation environment have emerged as a key research direction for next-generation wireless communication. Currently, existing flexible antenna architectures mainly include reconfigurable intelligent surfaces (RISs) [1], [2], fluid-antenna systems [3], [4], and movable antennas [5], [6]. The common underlying principle of these technologies is to actively reshape the end-to-end channel by dynamically adjusting the antenna position, radiation aperture, or electromagnetic response characteristics [7], thereby achieving performance gains. For instance, RISs create high-quality communication links by programmable manipulation of the electromagnetic environment to reconfigure signal propagation paths; fluid antennas rely on changes in the shape or position of the liquid medium for flexible reconfiguration; while movable antennas achieve physical migration within a confined space to find a superior channel state. By breaking the fixed-position and fixed-structure constraints of conventional antennas, the above aforementioned schemes show clear promise in improving channel quality and throughput.Although the flexible antenna system introduces an additional degree of freedom for wireless environment control, the flexible antenna system still faces two basic challenges. First of all, the adjustable range of the flexible antenna system is limited to the wavelength scale, so it is difficult to achieve the substantial enhancement of the [8] of the visual distance (LoS) link. Secondly, due to the complexity of manufacturing materials and structures, cost factors greatly limit the feasibility and popularity of flexible antenna systems in actual deployment.

To address the aforementioned limitations, pinching-antenna systems (PASS) has recently been proposed and garnered significant research attention. PASS leverages low-cost and readily available dielectric materials (e.g., plastic clips) that can be flexibly attached at arbitrary positions on a dielectric waveguide, dynamically creating new LoS links and thereby enhancing existing channel gains [9], [10]. Owing to its high deployment flexibility,low implementation cost, and reduced sensitivity to free-space path loss, PASS exhibits strong potential in wide-area coverage, robust communication in complex environments, and sustainable network deployment.

Furthermore, in scenarios oriented toward multi-user services, the architectural advantages of PASS become even more pronounced. In multi-user scenarios within the PASS, the system supports non-orthogonal multiple access (NOMA) technology, attributed to the superposition property of the dielectric waveguide signal [11]. This characteristic allows multiple user signals to be efficiently superimposed within the same resource block, thereby significantly improving spectral efficiency. However, most of the existing studies on PASS have not used this technology [12]. And with the increase in the number of users, the confidentiality performance of the system is also particularly important. For the scenario

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of PASS assisting multiple users, [13] does not consider the confidentiality performance of the system.

In order to solve the above problems, we have proposed a NOMA-assisted PASS, considering the maximization of the system confidentiality rate under the existence of multiple eavesdroppers and multi-users. However, due to the strong coupling of the transmission power and the clamping position of the antenna in this problem, it is not convex and difficult to deal with it directly. Therefore, to adress this problem, we have proposed an alternate optimization framework. Specifically, we decompose the original problem into two sub-problems, the optimized power distribution factor and the antenna pinching position. We use the successive convex approximation (SCA) algorithm to solve the power distribution factor sub-problem, and then use the particle swarm optimization (PSO) algorithm to solve the antenna locator problem. The simulation results show that the proposed optimization algorithm has good convergence, and also shows a significant enhancement to the confidentiality performance of the system.

II. System Model And Problem Formulation

A. System Model

We consider a NOMA-assisted PASS network, where a single PA transmits data to N users in the presence of K eavesdroppers. The PA is situated on a dielectric waveguide with a fixed height d and length L, running parallel to the x-axis, and can be dynamically activated at any point on the waveguide. We assume the users are randomly distributed within a rectangular service area in the x-y plane with dimensions D_x and D_y. In order to provide a clear description of the system geometry, a three-dimensional Cartesian coordinate system is utilized [16]. Denote the position of the PA by $\Phi^{\mathrm{Ant}} = \left(x^{\mathrm{Ant}}, 0, d\right)$, the i-th user and the k-th eavesdropper by $\Phi_i = (x_i, y_i, 0)$, $\Phi_{e,k} = (x_{e,k}, y_{e,k}, 0)$, respectively, where $x_i, x_{e,k} \in [0, D_x]$ and $y_i, y_{e,k} \in [-D_y/2, D_y/2]$. As in [17], we model the user channels using the free-space path loss model, considering only the LoS dominant path[1]. Thus, the channel power gain from the PA to the i-th user is given by [17]:

$$h_i = \frac{c^2}{16\pi^2 f_c^2 \left|\Phi_i - \Phi^{\mathrm{Ant}}\right|^2}. \tag{1}$$

where f_c and c represent the carrier frequency and the speed of light, respectively.

Similarly, the channel power gain from the PA to the k-th eavesdropper is given by:

$$h_{e,k} = \frac{c^2}{16\pi^2 f_c^2 \left|\Phi_{e,k} - \Phi^{\mathrm{Ant}}\right|^2}. \tag{2}$$

Since the PA employs the NOMA protocol, the downlink transmitted signal is a linear superposition of N data streams with distinct power allocation factors. Thus, the downlink NOMA signal can be expressed as:

$$x = \sum_{i=1}^{N} \sqrt{\alpha_i P} x_i, \tag{3}$$

where P is the total transmit power of the PA, x_i is the original information transmitted by the PA to the i-th user, and α_i is power allocation coefficient for the PA. Considering the total power constraint, we have $\sum\limits_{i=1}^{N} \alpha_i = 1$, with $0 \leq \alpha_i \leq 1$.

Consequently, the received signal at the i-th user is given by:

$$y_i = \sqrt{h_i} x + n_i = \sqrt{h_i} \sum_{i=1}^{N} \sqrt{\alpha_i P} x_i + n_i, \tag{4}$$

where $n_i \sim (0, \sigma^2)$ is the additive white Gaussian noise. Likewise, the received signal at the k-th eavesdropper is given by:

$$y_{e,k} = \sqrt{h_{e,k}} x + n_{e,k} = \sqrt{h_{e,k}} \sum_{i=1}^{N} \sqrt{\alpha_i P} x_i + n_{e,k}. \tag{5}$$

Without loss of generality, we assume that the channel power gains are sorted in ascending order with respect to the index, i.e., $|h_1|^2 \leq |h_2|^2 \leq \ldots \leq |h_N|^2$. Correspondingly, based on the NOMA protocol, the power allocation factors are allocated in descending order with respect to the index, i.e., $\alpha_1 \geq \alpha_2 \geq \ldots \geq \alpha_N$.

B. Problem Formulation

According to the NOMA protocol, the signal-to-interference-plus-noise ratio (SINR) at the i-th user from the PA can be expressed as:

$$\begin{aligned} \gamma_i &= \frac{\alpha_i P h_i}{\sum\limits_{j=i+1}^{N} \alpha_j P h_i + \sigma^2} \\ &= \frac{\alpha_i}{U_i + \frac{\sigma^2}{P h_i}}, \end{aligned} \tag{6}$$

where $U_i = \sum\limits_{j=i+1}^{N} \alpha_j$. The achievable transmission rate at the i-th user is given by:

$$R_i = \log_2\left(1 + \gamma_i\right). \tag{7}$$

Assume that the eavesdroppers have no prior knowledge of the decoding order, the SINR of the i-th user at the k-th eavesdropper is given by:

$$\begin{aligned} \gamma_{e,k}^{i} &= \frac{\alpha_i P h_{e,k}}{\sum\limits_{j \neq k} \alpha_j P h_{e,k} + \sigma_{e,k}^2} \\ &= \frac{\alpha_i}{U_e + \frac{\sigma_{e,k}^2}{P h_{e,k}}}, \end{aligned} \tag{8}$$

where $U_e = \sum\limits_{j \neq i} \alpha_j$. The achievable transmission rate at the k-th eavesdropper is given by:

$$R_{e,k}^{i} = \log_2\left(1 + \gamma_{e,k}^{i}\right). \tag{9}$$

[1]Due to the PASS has high deployment flexibility, low implementation cost and reduced sensitivity to free space path loss, the visual distance (LoS) component has an overwhelming advantage over scattered and non-visual distance (NLoS) paths in an environment with clear transmission links.

Therefore, based on the secrecy communication model, the achievable secrecy rate of the i-th user is given by:

$$\begin{aligned} S_i &= \sum_{k=1}^{K} \left(R_i - R_{e,k}^i\right)^+ \\ &= \sum_{k=1}^{K} \left[\log_2 \frac{(1+\gamma_i)}{(1+\gamma_{e,k}^i)}\right]^+ \\ &= \sum_{k=1}^{K} \left[\begin{array}{l} \log_2 \left(1+\frac{\alpha_i}{U_k+w_i}\right) \\ +\log_2(U_e+w_e) - \log_2(1+w_e) \end{array}\right]^+ \end{aligned} \tag{10}$$

where

$$w_i = \frac{\sigma^2}{h_i P}, w_e = \frac{\sigma^2}{h_{e,k} P}. \tag{11}$$

The objective of this work is to maximize the total secrecy sum-rate of all users by jointly optimizing the power allocation and the PA position. To guarantee the quality of service (QoS) for all users, we assume that each user must satisfy a minimum rate requirement, i.e.

$$R_i \geq R_i^{th}. \tag{12}$$

The optimization problem can be formulated as:

$$\begin{aligned} \text{(P0)}: \quad & \max_{\alpha_i, x^{\text{Ant}}} \sum_{i=1}^{N} S_i & \text{(13a)} \\ & \text{s.t.} \quad 0 \leq \alpha_i \leq 1, & \text{(13b)} \\ & \sum_{i=1}^{N} \alpha_i = 1, & \text{(13c)} \\ & R_i \geq R_i^{\text{th}}, & \text{(13d)} \\ & x^{\text{Pin}} \in [0, L]. & \text{(13e)} \end{aligned}$$

III. Proposed Solution

Problem (P0) is difficult to solve due to the coupling between transmit-power variables and the PA position, which makes direct optimization intractable. To handle this issue, we employ an AO framework that divides the problem into power-allocation and PA-positioning subproblems, solved iteratively to reach convergence.

A. Power Allocation under Given Pinching Antenna Position

For a fixed PA position, the power allocation subproblem can be re-written as:

$$\begin{aligned} \text{(P1)}: \quad & \max_{\alpha_i, U_i, U_e} \sum_{i=1}^{N} \log_2 \left(1 + \frac{\alpha_i}{U_i + w_i}\right) + \log_2(U_e + w_e) \\ & \qquad - \log_2(1+w_e) & \text{(14a)} \\ & \text{s.t.} \quad 0 \leq \alpha_i \leq 1, & \text{(14b)} \\ & \sum_{i=1}^{N} \alpha_i = 1, & \text{(14c)} \\ & \gamma_i \geq \gamma_i^{\text{th}}. & \text{(14d)} \end{aligned}$$

In the above constraint, since $\gamma_i^{\text{th}} = 2^{R_i^{\text{th}}} - 1$, we have $\alpha_i \geq \gamma_i^{\text{th}}(U_i + w_i)$. Observing (14a), it can be easily demonstrated that the objective function is concave with respect to α_i and convex with respect to U_e. Thus, the objective function in (14a) is non-convex, which makes it difficult to solve. To resolve this non-convexity, this work employs the successive convex optimization technique to provide a tightly approximated solution. By applying the first-order Taylor expansion at a prior feasible solution, the non-convex function is converted into a linear function. To perform the successive convex optimization, we introduce a set of auxiliary variables r_k [18], and the optimization problem (P1) is reformulated as:

$$\begin{aligned} \text{(P2)}: \quad & \max_{\alpha_i, U_i, U_e, r_k} \sum_{i=1}^{N} \log_2(U_i + w_i + \alpha_i) + \log_2(U_e + w_e) \\ & \qquad - \log_2 r_k - \log_2(1+w_e) & \text{(15a)} \\ & \text{s.t.} \quad 0 \leq \alpha_i \leq 1, & \text{(15b)} \\ & \sum_{i=1}^{N} \alpha_i = 1, & \text{(15c)} \\ & \alpha_i \geq \gamma_i^{\text{th}}(U_i + w_i), & \text{(15d)} \\ & U_i + w_i \leq r_k. & \text{(15e)} \end{aligned}$$

Since the objective function in (15a) is convex with respect to r_k, we can obtain the following lower bound $\tilde{r}_k^r$ by fixing the r-th iteration result.

$$\begin{aligned} S_i' \geq & \log_2(U_i + w_i + \alpha_i) + \log_2(U_e + w_e) \\ & - \log_2(1+w_e) - \log_2 \tilde{r}_k^r - \frac{1}{\ln 2 \cdot \tilde{r}_k^r}(r_k - \tilde{r}_k^r) \\ = & S_k^{\text{lb}}. \end{aligned} \tag{16}$$

Therefore, the optimization problem in (15) can be reformulated as:

$$\begin{aligned} \text{(P3)}: \quad & \max_{\alpha_i, U_i, U_e, r_k} \sum_{i=1}^{N} S_i^{\text{lb}} & \text{(17a)} \\ & \text{s.t.} \quad 0 \leq \alpha_i \leq 1, & \text{(17b)} \\ & \sum_{i=1}^{N} \alpha_i = 1, & \text{(17c)} \\ & \alpha_i \geq \gamma_i^{\text{th}}(U_i + w_i), & \text{(17d)} \\ & U_i + w_i \leq r_k, & \text{(17e)} \end{aligned}$$

It can be seen that the objective function in (17a) is concave with respect to α_i, U_i, and U_e, and affine with respect to r_k. Meanwhile, all constraints in P3 are affine. Consequently, the optimization problem in P3 is convex, which can be efficiently solved using standard convex optimization tools (e.g., CVX).

B. Antenna Position Optimization under Given Power Allocation

In this section, given a fixed power allocation factor, the objective is to maximize the total secrecy sum-rate by optimizing the PA position. By expanding the definitions of h_i and $h_{e,k}$, we have:

$$h_i = \frac{c^2/16\pi^2 f_c^2}{|x^{\text{Ant}} - x_i|^2 + y_i^2 + d^2}, \tag{18}$$

$$h_{e,k} = \frac{c^2/16\pi^2 f_c^2}{|x^{\mathrm{Ant}} - x_{e,k}|^2 + y_{e,k}^2 + d^2}. \tag{19}$$

Based on the original optimization problem (P0), by substituting the expanded expressions of h_i and $h_{e,k}$ into (17a), the problem of determining the optimal PA position to maximize the total secrecy sum-rate can be expressed as:

$$\text{(P4)}: \quad \max_{x^{\mathrm{Ant}}} \quad \sum_{i=1}^{N} S_i \tag{20a}$$

$$\text{s.t.} \quad x^{\mathrm{Ant}} \in [0, L]. \tag{20b}$$

In optimization problem (P4), (20a) represents a generalized bell-shaped membership function, characterized by its corresponding bell curve, which is inherently non-convex. Existing literature [10] has proven that the sum of such functions can lead to multiple local optima. The PSO algorithm represents an effective solution method to this non-convex optimization problem. Notably, the solution obtained via PSO in [10] shows high agreement with the results from exhaustive search, thereby validating the effectiveness of the method. In this paper, we employ the PSO algorithm to optimize the pinching antenna position as in [10].

C. Alternating Optimization until Convergence

To guarantee convergence, an AO framework is adopted, where the power allocation and PA positioning subproblems are solved alternately. The power allocation subproblem is optimally solved via SCA using CVX, ensuring a non-decreasing secrecy sum rate across iterations, while the PA positioning subproblem is handled by PSO with a termination rule that retains the previous position if no performance improvement is observed. In terms of computational complexity, each AO iteration incurs $\mathcal{O}(I_{\mathrm{sca}}N^3)$ for power allocation and $\mathcal{O}(I_{\mathrm{pso}}SNK)$ for PA positioning, where I_{sca} denotes the maximum number of inner iterations of the SCA algorithm, S represents the swarm size in the PSO algorithm, I_{pso} is the maximum number of PSO iterations. N and K denote the numbers of legitimate users and eavesdroppers, respectively. Therefore, after T AO iterations, the overall computational complexity is $\mathcal{O}\big(T\big(I_{\mathrm{sca}}N^3 + I_{\mathrm{pso}}SNK\big)\big)$.

IV. Numerical Results

In this section, we present simulation results to validate the effectiveness and convergence of the proposed algorithm. The following default parameters are employed for the simulations: the antenna height is $d = 3$ m, the carrier frequency is $f_c = 28$ GHz, and the noise power is $\sigma^2 = -97$ dBm. Legitimate users and eavesdroppers are randomly deployed in a service area defined by $D_x = 60$ m and $D_y = 20$ m. Additionally, the length of the dielectric waveguide is configured to match the horizontal span of the service area, i.e., $L = D_x$.

Fig.1 illustrates the convergence performance of the CVX algorithm under different transmit power constraints. The simulation is configured with $N = 4$ legitimate users and $K = 2$ eavesdroppers. We observe that the total secrecy sum rate of the system monotonically increases with the number of iterations and eventually converges to a stable value. This strongly verifies the effectiveness and robustness

Algorithm 1 Overall AO algorithm for joint optimization of α_i, $\mathrm{x}^{\mathrm{Ant}}$

1: Initial PA positions $\mathrm{x}^{\mathrm{Ant}(0)}$, initial power allocation factor $\alpha^{i(0)}$, maximum iterations r.
2: Set $r = 0$. Evaluate initial equivalent channels $h_i^{(0)}$, $h_{e,k}^{(0)}$ and initial objective $S_i^{(0)}$.
3: **repeat**
4: Update the channel gain h_i,$h_{e,k}$ by (1),(2);
5: Update the interference w_i,e by (12);
6: Update the power allocation factor α_i using CVX algorithm;
7: Update PA position $\mathrm{x}^{\mathrm{Ant}}$ using PSO algorithm;
8: Evaluate new objective $S_i^{(r+1)}$;
9: $r = r + 1$.
10: **until** $r \geq r_{\max}$.
11: **return** $h_i^{(r)}, h_{e,k}^{(r)}, \mathrm{x}^{\mathrm{Ant}}$.

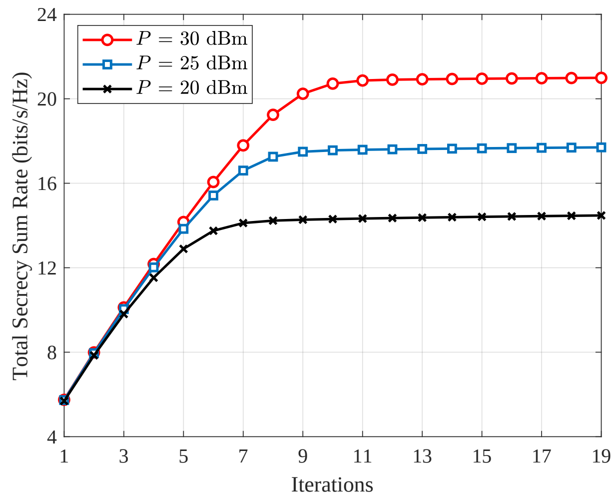

Fig. 1. Convergence of Algorithm 1 with respect to iteration r.

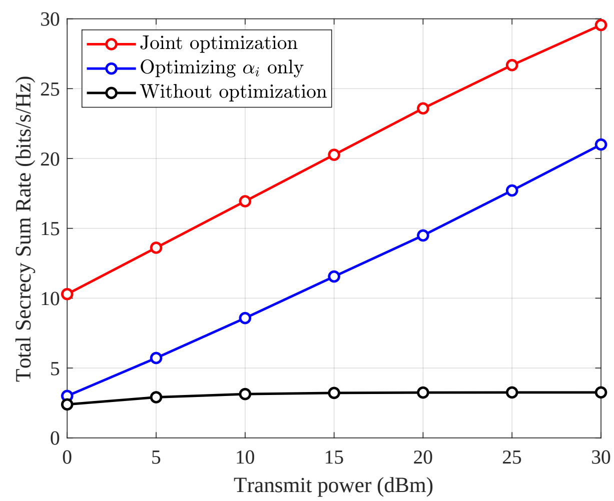

Fig. 2. Comparison of the relationship between system total secrecy sum rate and transmit power under different optimization schemes.

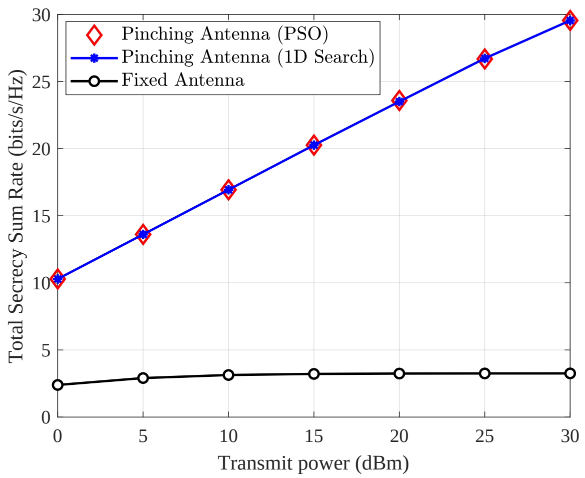

Fig. 3. Comparison of the relationship between system total secrecy and rate and transmit power under different antenna configuration optimization schemes.

of the proposed iterative algorithm, which is based on SCA, for solving the non-convex power allocation subproblem. Furthermore, as the transmit power increases, the system achieves a higher final total secrecy sum rate. Based on this convergence characteristic, the maximum number of iterations for the algorithm is set to 30 in the subsequent simulations to balance computational complexity and convergence accuracy.

Fig.2 illustrates the total secrecy sum rate for all schemes monotonically increases with the growth of transmit power. Critically, the proposed joint optimization scheme (CVX+PSO) exhibits the superior performance, significantly outperforming the fixed-antenna scheme with CVX. This superiority is attributed to the additional spatial degrees of freedom provided by the PASS, which dynamically adjusts the antenna position to reshape the channel environment. The scheme with arbitrarily predefined antenna positions and power allocation coefficients, without any optimization, yields the lowest performance, underscoring the necessity of the joint optimization design of power allocation and dynamic antenna positioning for enhancing system security performance.

Fig.3 presents the performance curve of the proposed PSO algorithm closely aligns with that obtained from the more complex 1D Search benchmark. This congruence validates that the PSO algorithm efficiently converges to a near-globally optimal solution for the non-convex antenna placement subproblem with significantly lower computational overhead. Consequently, the PSO scheme achieves a favorable trade-off between computational complexity and solution accuracy while maximizing the total secrecy sum rate, thus enhancing its practical viability.

V. Conclusion

In this paper, we studied the secrecy sum-rate maximization problem for a NOMA-assisted PASS subject to legitimate-channel achievable-rate constraints. The problem was formulated as a challenging non-convex joint optimization of user transmit powers and the dynamic PA position. To address this, an AO framework was adopted. In particular, the power-allocation subproblem was efficiently solved using the SCA method, while the antenna-positioning subproblem was effectively handled via a PSO algorithm. Simulation results corroborate the analytical insights, showing that the proposed joint CVX+PSO scheme achieves substantial secrecy-sum-rate gains over the conventional fixed-antenna baseline. Moreover, the iterative SCA procedure demonstrates stable convergence behavior, and the PSO-based positioning strategy attains near-optimal performance relative to exhaustive search, thereby validating the robustness and effectiveness of the proposed approach.

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